Structural certificates for the A_9 and S_9 tables

Private L054 finite application

Theorem 1. *Relative to the acquired complete counts of ambient subgroup classes, exact local maximal subgroup interfaces and earlier structural registry, all 223 rows of the current A_9 table and all 554 rows of the current S_9 table have the displayed structural descriptions, with the explicit Q_{20} presentation below. These 777 descriptions assert actual decompositions, not uniqueness of an isomorphism type from a label.*

Source syntax. The current companion PDFs are attributed to Thomas Connor and Dimitri Leemans. All ten A_9 and 27 S_9 native pages were directly read. Historical identity with the 2013 source files is not asserted. The 2015 article's left-association convention is used unless the native source has parentheses. In particular, S_9 row 99 is $D_8 \times (D_{10} \cdot 2)$, not $(D_8 \times D_{10}) \cdot 2$. Row 157's $D_8 \times (D_{10})$ has redundant parentheses around a single atom. Row 22 preserves the dot in $((((A_4 \times 2) \times 2) : S_3) \cdot 2) : 2$. Powers, dots, colons and crosses are retained exactly. An unsigned extraspecial atom asserts no plus or minus type.

Recovering actual embeddings. The retained S_9 group display contains five nonliteral constructor expressions, in addition to point-set symmetric and alternating names. We do not assume that a display retained the original producer's embedding. The conventional constructor outputs, including a proposed replacement of printed $L(3,2)$ by $\mathrm{PSL}(3,2)$, are only candidate groups. Every candidate is exported as actual ordered permutation generators. The A_9 point-set names are treated in the same way.

Let the acquired complete number of subgroup conjugacy classes of G be c. If c candidate subgroups belong to G and are pairwise nonconjugate in G, their classes exhaust the subgroup classes of G. This elementary counting argument was already used in the accepted A_7, S_7, A_8, S_8 application. Here the unchanged robust local-weight kernel checks these conditions for all 223 and 554 candidates, respectively. The actual ambient models are checked against the natural alternating and symmetric groups on nine points. Thus no interpretation of a printed proper-group name is a completeness premise.

For each candidate H, the acquired local interface supplies a complete covering list of genuine maximal subgroups of a checked faithful smaller action. Literal conjugacy tests remove repetitions, and inverse transport returns actual maximal subgroups of H. Fusion into the validated ambient roster is by actual conjugacy. Each local class contributes $[H : N_H(K)]$ children; contributions with the same ambient class are added. If the ambient lengths of the parent and child classes are c_H, c_K, double counting conjugate incidences gives $b = ac_H/c_K$ parents above each child. Both integrality and the bounds $a \le c_K$, $b \le c_H$ are checked. This is the accepted robust weighting theorem and kernel, not an assumption that a software class list has no repeated classes.

The saved source bijections are tested anew against every row order, class length and both directions of every edge and nonedge. The new arrays have 722 and 2267 weighted edges, and all 356,645 ordered source pairs agree. These checks establish the source assignment of the actual candidate groups. No ambient subgroup enumeration is repeated.

Registry and structural proof. There are 189 acquired concrete registry models and two elementary natural models, A_9, S_9. A full cycle and a transposition generate the symmetric model; the alternating model uses $(1\,2\,i)$ for $3 \le i \le 9$. The literal closures have orders 9! and 9!/2, and the latter generators are even, proving the two named identifications.

The application has 82 new models: 40 split, 32 direct, nine nonsplit, and one dicyclic model. Factor references are acyclic. A structural reuse is always an explicit bijection fixing identity and preserving right multiplication by every ordered generator. Induction on generator words makes it an isomorphism; this is the acquired MC.3 criterion. All 930 such maps, comprising 4,038,442 generator-edge equations, are checked.

For a split model, the verifier checks $N \triangleleft H$, identified factors N, K, trivial intersection and $|N||K| = |H|$. Thus $H = NK$. A colon requires a noncommuting pair of factor generators. A cross requires all factor generators to commute, hence elementwise commutation. These are actual subgroup and multiplication checks.

Six new nonsplit models have prime quotient, and every outside element is checked against the requisite prime-order condition: 394 tests in all. Three models have quotient S_3. Fixed quotient generators xN, yN satisfy $x^2, y^3, (xy)^2 \in N$, and six explicit distinct cosets prove the quotient presentation. A complement exists exactly when some lifts $a \in Nx, b \in Ny$ obey $a^2 = b^3 = (ab)^2 = 1$. The relations then give a surjection from S_3 onto the generated lift group, while its quotient surjects onto S_3, forcing order six and a complement. Conversely a complement supplies such lifts. All 1940 pairs fail. These are the previously accepted exhaustive quotient-lift criteria; a failed software complement search is never evidence of nonsplitting.

The precise Q_{20} convention. The source does not separately define this notation. As with the acquired Q_{12} convention, we identify the dicyclic group by the presentation

$$Q_{20} := \langle x, y \mid x^{10} = 1,\ y^2 = x^5,\ y^{-1}xy = x^{-1} \rangle.$$

Every word reduces to x^i or $x^i y$, $0 \le i < 10$, so the presentation has at most twenty elements. The actual new model supplies generators of an exactly twenty-element group, with $|x| = 10$, satisfying these relations. Hence the presented group and the actual model are isomorphic. The installed GAP Reference Manual 50.1-8–9 defines this dicyclic presentation, and reserves generalized quaternion for the 2-power-order case. We retain the explicit nomenclature qualification rather than calling order twenty a quaternion 2-group. Source S_9 row 379 and its occurrence inside row 260's $Q_{20} : 2$ have actual maps to this model.

Validation and limits. The bounded witness pass returned all 777 rows in 58.99 seconds. It proposed witnesses only. The final replay closes and verifies every literal finite group, map, decomposition and source join, with identical ordinary and optimized reports. The embedding revalidations took 9.78 and 29.51 seconds; their exact clean receipts and all completed parent outputs are retained and acquired without an unchanged repeat.

Independent regular dicyclic actions of orders 12 and 20 test the new criterion. Wrong rotations and lifts, genuine dihedral and cyclic groups of order twenty, malformed source parentheses, wrong row generators and assignments, changed dots and orders, nonbijective source maps, duplicate edges and incorrect class lengths or weights are rejected. The accepted transport and extension controls remain acquired. No canonical uniqueness of structural notation, historical PDF-byte identity, whole-source admission, question resolution, formalization or measurement is claimed.